

PATIENT NAME: Jane Doe	SAMPLE NUMBER: S-123456
DATE OF BIRTH: 1990-01-01	PHYSICIAN: Dr. Smith
GESTATIONAL AGE: 12 Weeks 3 Days	HOSPITAL / CLINIC: General Hospital
PREGNANCY TYPE: Singleton	COLLECTION DATE: 2023-10-01
MRN: MRN-123456	REPORT DATE: 2023-10-05
INDICATION: Advanced Maternal Age	ORDER ID: ORD-7890

Overall Result: Low Risk

Predicted Fetal Sex Female

Non-invasive prenatal screening for common fetal chromosome abnormalities.

Fetal Fraction 10.5%

CHROMOSOMAL ANEUPLOIDIES

CONDITION	RESULT	CONDITION	RESULT
Trisomy 21 (Down Syndrome)	Low Risk	Trisomy 9	Low Risk
Trisomy 18 (Edwards Syndrome)	Low Risk	Trisomy 16	Low Risk
Trisomy 13 (Patau Syndrome)	Low Risk	Trisomy 22	Low Risk
All Other Autosomal Trisomy	Low Risk		
XO (Turner Syndrome)	Low Risk	XXX (Trisomy X)	Low Risk
XXY (Klinefelter Syndrome)	Low Risk	YYY (Jacob Syndrome)	Low Risk

MICRODELETIONS & DUPLICATIONS

CONDITION	RESULT	CONDITION	RESULT
1p36 deletion syndrome	Low Risk	Wolf-Hirschhorn syndrome	Low Risk
Williams-Beuren syndrome	Low Risk	Prader-willi/Angelman syndrome	Low Risk
Cri Du Chat syndrome	Low Risk	Jacobsen syndrome	Low Risk
DiGeorge syndrome (22q11.2)	Low Risk	2q33.1 deletion syndrome	Low Risk
Other microdeletions/duplications	Low Risk		

INTERPRETATION

This screening test indicates a low risk for the common aneuploidies and microdeletion syndromes evaluated. No evidence of fetal chromosome aneuploidy or the targeted microdeletion/duplication syndromes was detected. Fetal fraction was adequate for analysis. This is a screening test, not diagnostic; confirmatory testing is recommended if clinical suspicion remains.

Non-Invasive Prenatal Screening

Performance & Methods



PERFORMANCE METRICS

CONDITION	SENSITIVITY	SPECIFICITY	PPV
Trisomy 21	99.72%	>99%	98.72%
Trisomy 18	99.04%	>99%	96.49%
Trisomy 13	98.84%	>99%	85.71%
Monosomy X	>99%	>99%	76%
XXX	>99%	>99%	>99%
XXY	>99%	>99%	>99%
XYY	>99%	>99%	>99%
T9, T16, T22	>99%	>99%	50–99%
All Other Trisomy	>99%	>99%	NA
Microdeletions / duplications	20–99%	>99%	NA

METHODS AND LIMITATIONS

Limitations: This test is designed to screen for fetal chromosome abnormalities in singleton pregnancies that are at least 10 weeks gestation, as estimated by last menstrual period. Microdeletion sensitivity is dependent on size and fetal fraction; the minimum detectable size is 0.5 MB at 4% fetal fraction. This test does not rule out other chromosomal or subchromosomal abnormalities, birth defects, or open neural tube defects. A negative result does not exclude all chromosomal abnormalities. There is a small chance results may reflect placental or maternal chromosomal changes rather than the fetus.

Methods: Low-pass whole-genome sequencing from extracted cell-free DNA. Sequencing data validated in-house. PPV performance calculated from amniocentesis and QF-PCR results. Where amniocentesis samples were insufficient, sensitivity was estimated from PPV.

References:

1. Anthony R et al., Noninvasive prenatal screening for fetal aneuploidy 2016 update: a position statement of the American College of Medical Genetics and Genomics. Genet Med 2016; 13:1056-65.
2. Bianchi DW et al., DNA sequencing versus standard prenatal aneuploidy screening. N Engl J Med. 2014; 370:799-808.
3. Lo YMD, Chiu RWK, Noninvasive prenatal diagnosis of fetal chromosomal aneuploidies by maternal plasma nucleic acid analysis. Clin Chem, 2008.
4. Chen M. et al., Validation of fetal DNA fraction estimation and its application in NIPT for aneuploidy detection in multiple pregnancies. Prenat Diagn, 2019. 39(13): 1273-1282.

MICRODELETIONS / DUPLICATIONS SYNDROMES EVALUATED

REGION / SYNDROME	REGION / SYNDROME	REGION / SYNDROME	REGION / SYNDROME
1p36 terminal region (includes GABRD) gain	Partial trisomy 5p	Partial trisomy 12p	17q11.2 recurrent region (includes NF1) gain
1q21.1 recurrent region (distal, BP3-BP4) (includes GJA5) loss	Partial monosomy 5q	Partial trisomy 12q	17q11.2 recurrent region (includes NF1) loss
1q21.1 recurrent region (distal, BP3-BP4) (includes GJA5) gain	Partial trisomy 5q	Monosomy 13q14	17q12 recurrent (RCAD syndrome) region (includes HNF1B) gain
1q43-q44 terminal region (includes AKT3)	6q24 region (includes PLAGL1)	Monosomy 13q21-qter	17q12 recurrent (RCAD syndrome) region (includes HNF1B) loss
Partial monosomy 1p	Partial trisomy 6p	Trisomy 13cen-q14	17q21.31 recurrent region (includes KANSL1)
Monosomy 1q21-q32	Partial monosomy 6q	Trisomy 13q21-qter	17q23.1-q23.2 recurrent region (includes TBX2, TBX4) gain
Monosomy 1q42-qter	Partial trisomy 6q	14q11.2 region including CHD8 and SUPT16H	17q23.1-q23.2 recurrent region (includes TBX2, TBX4) loss
Trisomy 1q23-qter	7p22.1 region (includes ACTB)	DLK1-MEG3 Intergenic Region	SOX9 upstream enhancer region
2p15-p16.1 region (includes BCL11A)	7q11.23 recurrent (Williams-Beuren syndrome) region (includes ELN) gain	Partial trisomy 14	Trisomy 17pter-q21
2p24.3 MYCN-DDX1 duplication region	7q11.23 recurrent distal region (includes HIP1, YWHAG)	Trisomy 14q24-qter	Partial monosomy 18p
2q11.2 recurrent region (includes ARID5A, TMEM127)	7q36.3 ZRS (SHH cis-regulatory) duplication region (within LMBR1 intron 5)	15q11.2-q13 recurrent (PWS/AS) region (Class I, BP1-BP3 and Class II, BP2-BP3) gain	Partial tetrasomy 18p
2q13 recurrent region (distal) (includes BCL2L11) gain	Partial monosomy 7p	15q13.3 recurrent region (BP4-BP5) (includes CHRNA7)	Trisomy 18pter-q12
2q13 recurrent region (distal) (includes BCL2L11) loss	Partial trisomy 7p	15q13.3 recurrent region (D-CHRNA7 to BP5) (includes CHRNA7, OTUD7A)	Monosomy 18q21-qter
Partial monosomy 2p	Monosomy 7q32-qter	15q24 recurrent region (LCR A-LCR C)	Trisomy 18q12-qter
Partial trisomy 2p	Trisomy 7q21-q31	15q24 recurrent region (LCR A-LCR D) (includes SIN3A)	Partial monosomy 20p
Partial monosomy 2q	Trisomy 7q32-qter	15q24 recurrent region (LCR C-LCR D) (includes SIN3A)	Trisomy 20p
Partial trisomy 2q	8p23.1 recurrent region (includes GATA4) gain	15q25.2 recurrent region (proximal, LCR-A/B-C) (includes RPS17)	Partial monosomy 21
3q24 Region (includes ZIC1)	8p23.1 recurrent region (includes GATA4) loss	Supernumerary inv dup (15)	22q11.2 recurrent (DGS) region (proximal, A-B, A-C, or A-D) (includes TBX1) gain
3q29 recurrent region (includes DLG1)	Partial trisomy 8q	Trisomy 15q22-qter	22q11.2 recurrent region (central, B-D or C-D) (includes CRKL)
Monosomy 3p11-p21	Partial monosomy 9p	16p11.2 recurrent region (distal, BP2-BP3) (includes SH2B1)	22q11.2 recurrent region (distal type I, D-E or D-F)
Monosomy 3p25-pter	Partial trisomy 9p	16p11.2 recurrent region (proximal, BP4-BP5) (includes TBX6) gain	22q11.2 recurrent region (distal type III, D-G, D-H) (includes SMARCB1)
Partial trisomy 3p	Partial monosomy 9q	16p11.2 recurrent region (proximal, BP4-BP5) (includes TBX6) loss	22q11.2 recurrent region (distal type III, E-H or F-H) (includes SMARCB1)
Partial monosomy 3q	Partial trisomy 9q	16p12.2 recurrent region (proximal) (includes EEF2K, CDR2)	22q11.2 recurrent region (distal type III, F-G) (includes SMARCB1)
Partial trisomy 3q	10q22.3-q23.2 recurrent region (includes BMPR1A)	16p13.11 recurrent region (BP1-BP3, BP2-BP3, or BP2-BP4) (includes MYH11) gain	22q11.21 recurrent (CES) region (includes CECR2)
Partial monosomy 3p	Partial monosomy 10p	16p13.11 recurrent region (BP1-BP3, BP2-BP3, or BP2-BP4) (includes MYH11) loss	Xp22.31 recurrent region (includes STS)
4p16.3 terminal (Wolf-Hirschhorn syndrome) region gain	Partial trisomy 10p	16p13.3 region (includes CREBBP)	Xp21.2 region (includes NROB1)
Partial monosomy 4p	Partial monosomy 10q	Partial trisomy 16p	Xp11.23 region (includes MAOA and MAOB)
Partial trisomy 4p	Partial trisomy 10q	Partial monosomy 16q	Xp11.22-p11.23 recurrent region (includes SHROOM4)
Monosomy 4q21-q31	11p11.2 (Potocki-Shaffer syndrome) region (includes ALX4, EXT2)	Partial trisomy 16q	Xp11.22 region (includes HUWE1)
Monosomy 4q31-qter	11p13 (WAGR syndrome) region	17p11.2 recurrent (SMS/PLS) region (includes RAI1) gain	Xq25 region (includes STAG2) gain
Partial trisomy 4q	11q13.2-q13.4 recurrent region (includes SHANK2, FGF3)	17p11.2 recurrent (SMS/PLS) region (includes RAI1) loss	Xq25 region (includes STAG2) loss
Partial monosomy 4p	Partial trisomy 11p	17p12 recurrent (HNPP/CMT1A) region (includes PMP22) gain	Xq28 recurrent region (includes GDI1)
5p15 terminal (Cri du chat syndrome) region gain	Partial monosomy 11q	17p12 recurrent (HNPP/CMT1A) region (includes PMP22) loss	Xq28 recurrent region (int22h1/int22h2-flanked) (includes RAB39B) gain
5q35 recurrent (Sotos syndrome) region (includes NSD1) gain	Partial trisomy 11q	17p13.3 (Miller-Dieker syndrome) region (includes YWHAE and PAFAH1B1) gain	Xq28 recurrent region (int22h1/int22h2-flanked) (includes RAB39B) loss
5q35 recurrent (Sotos syndrome) region (includes NSD1) loss	Partial monosomy 12p	17p13.3 (Miller-Dieker syndrome) region (includes YWHAE and PAFAH1B1) loss	Xq28 region (includes MECP2) gain
			Xq28 region (includes MECP2) loss